

Homework 2: Python Simulation Questions

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Overview

This report presents the Python simulation results for the simulation-based questions from Homework 2. The focus is on Questions 10, 11, and 17. Questions 10 and 11 compare different prediction rules using MAE, MAPE, and MARE. Question 17 empirically demonstrates that a regression model can achieve perfect training fit when it has enough linearly independent features.

Metric Definitions

The mean absolute error (MAE) is defined as

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |y_t - \hat{y}_t|.$$

MAE measures the average raw absolute prediction error, so it is expressed on the same scale as the response variable.

The mean absolute percentage error (MAPE) is defined as

$$\text{MAPE} = 100 \times \frac{1}{N} \sum_{t=1}^N \frac{|y_t - \hat{y}_t|}{|y_t|}.$$

MAPE measures error relative to the observed value. This means that errors when the observed value is small can have a large influence on the metric.

The mean absolute relative error (MARE) is defined as

$$\text{MARE} = 100 \times \frac{1}{N} \sum_{t=1}^N \frac{|y_t - \hat{y}_t|}{|\hat{y}_t|}.$$

MARE is similar to MAPE, but it divides by the predicted value instead of the observed value.

Question 10

Setup

The data were generated using a two-part Poisson mean function. For the first 25 time points, the expected value is constant. For the final 25 time points, the expected value increases exponentially. Two hypothetical prediction vectors were then compared:

- Model 1: shifted upward by 1.5 in both the constant and exponential portions.
- Model 2: correctly centered early, but with a slower exponential growth rate later.

Results

Table 1: Question 10 error metrics.

Model	MAE	MAPE
Model 1	3.471	52.904
Model 2	5.477	38.897

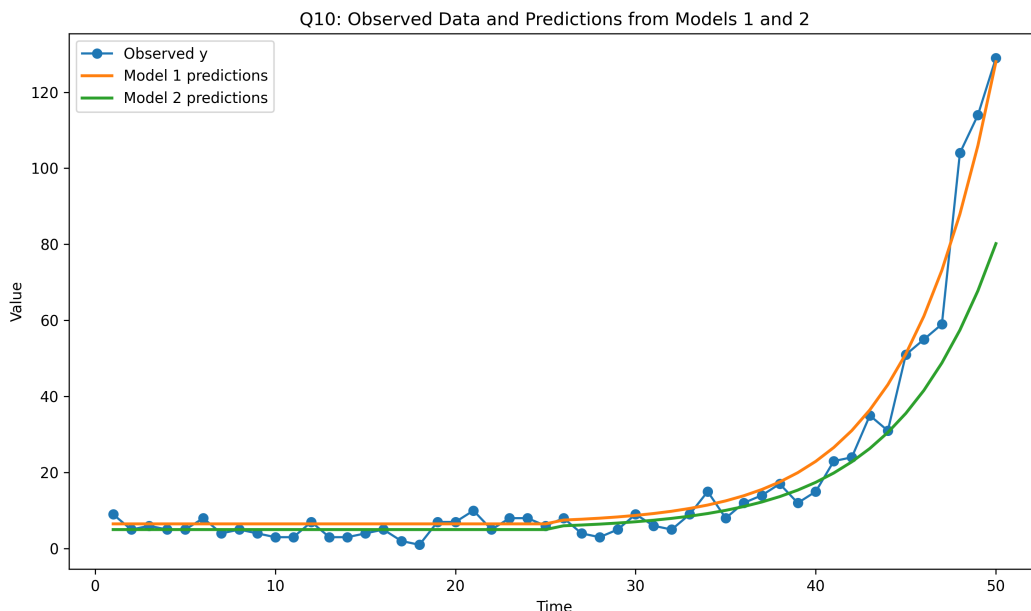


Figure 1: Observed data and predictions from Models 1 and 2.

Discussion

The two metrics favor different models. Model 1 has the lower MAE, which means that it has the smaller average absolute error on the original scale of the data. However, Model 2 has the lower MAPE, which means that it has the smaller average percentage error.

This difference occurs because MAE and MAPE emphasize different parts of the time series. MAE is sensitive to large raw errors, especially during the high-count exponential part of the series. MAPE divides each absolute error by the observed value, so errors during the low-count part of the series receive more weight. Model 1 is shifted upward in the first half of the series, so even though its raw errors are not very large, its percentage errors can be large when the observed counts are small. Therefore, Model 1 looks better by MAE, while Model 2 looks better by MAPE.

Question 11

Setup

A third prediction vector, Model 3, was created by starting with Model 2 and changing the exponential multiplier in the final 25 predictions from 0.18 to 0.22. This makes Model 3 grow faster than Model 2 in the second half of the series.

Results

Table 2: Question 11 error metrics.

Model	MAE	MAPE	MARE
Model 2	5.477	38.897	32.913
Model 3	8.157	48.049	33.276

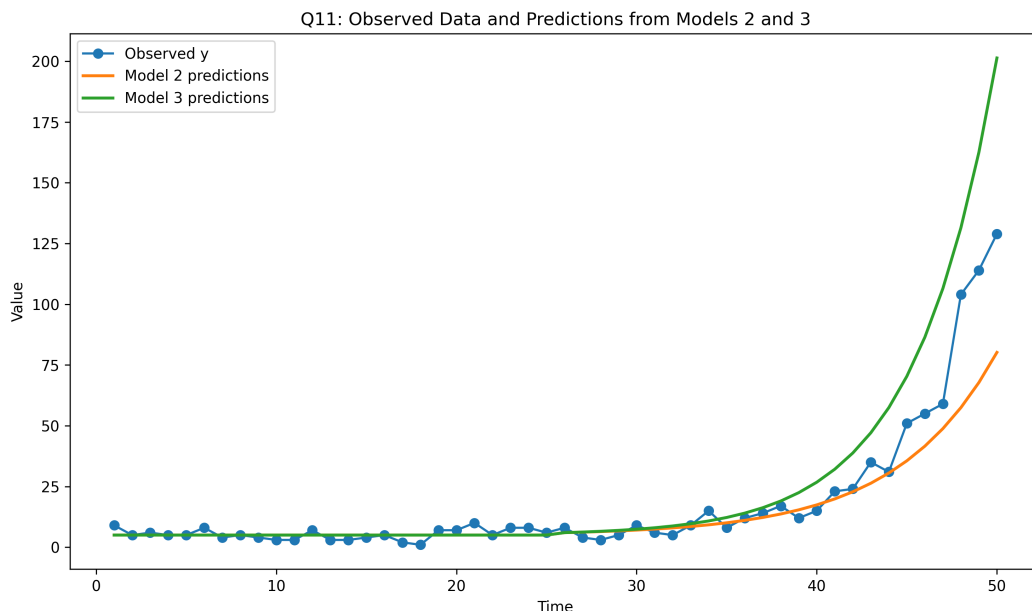


Figure 2: Observed data and predictions from Models 2 and 3.

Discussion

Model 3 performs worse than Model 2 according to both MAE and MAPE. This happens because the larger exponential multiplier causes Model 3 to grow too quickly in the second half of the time series. As a result, Model 3 tends to overshoot the observed values more strongly than Model 2.

The MARE comparison is more subtle. MARE divides the absolute error by the predicted value rather than by the observed value. Since Model 3 produces larger predictions in the second half of the series, the denominator in MARE is also larger. This can make large overpredictions look less severe than they would under MAPE. In this simulation, Model 3 is still slightly worse by MARE, but the gap between the two models is much smaller under MARE than under MAE or MAPE. This illustrates the main lesson of the question: the choice of error metric can meaningfully change how model performance is interpreted.

Question 17

Goal

Question 16 asks how many linearly independent features are needed to achieve perfect training accuracy. The answer is that, for n observations, we need enough linearly independent features to

span the n observed responses. In particular, if the feature matrix has rank n , then ordinary least squares can exactly interpolate the training responses, giving a training MSE of zero.

Question 17 asks for an empirical verification of this result. To verify it, I simulated $n = 50$ response values and used the 50×50 identity matrix as the feature matrix. Each observation therefore has its own linearly independent feature. I then fit a linear regression model with no intercept.

Results

The training mean squared error was

$$\text{Training MSE} = 0.000000000000.$$

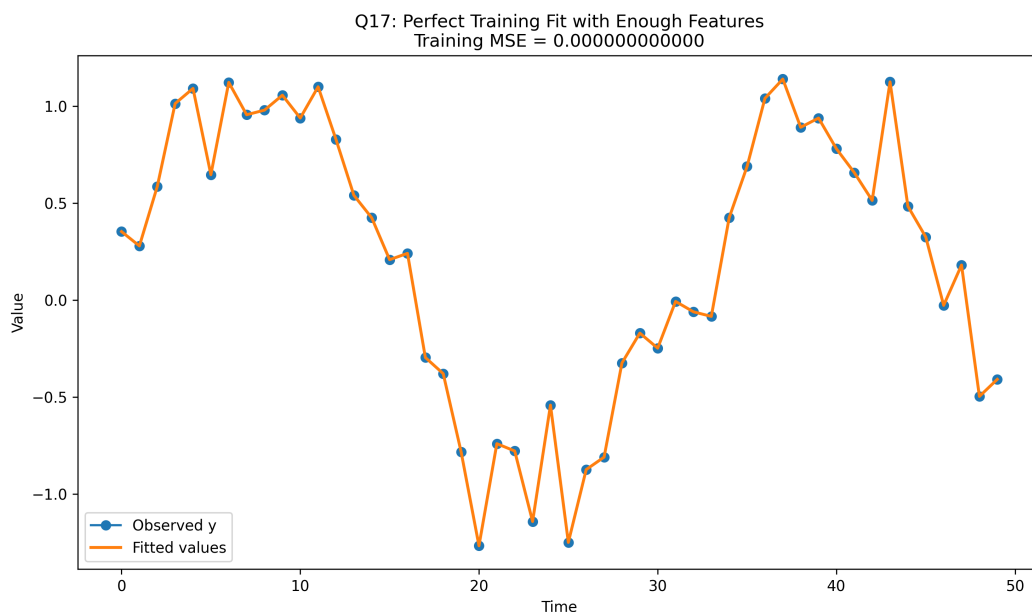


Figure 3: Observed data and fitted values using an identity-matrix feature design. The fitted values match the observations exactly.

Discussion

The fitted values match the observed values exactly because the identity matrix gives the model one linearly independent feature per observation. In this setup, the regression model can memorize the training data. This confirms empirically that with enough linearly independent features, the training MSE can be driven to zero.

However, this does not imply that the model would generalize well to new data. In fact, this is an example of overfitting: the model achieves perfect training performance by memorizing the sample rather than learning a stable predictive relationship.

Conclusion

The simulation results show that model comparisons depend strongly on the metric being used. In Question 10, MAE favors Model 1 while MAPE favors Model 2 because raw-scale errors and

percentage errors emphasize different parts of the data. In Question 11, Model 3 is worse than Model 2 by MAE and MAPE, but MARE makes the difference appear smaller because it scales errors by the predicted values. Finally, Question 17 shows that adding enough linearly independent features can produce perfect training fit, but this should be interpreted as memorization rather than reliable forecasting performance.